Blockchain-Based Professional Credential Verification: A Cardano Smart Contract Architecture for License Management and Digital Signatures

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***Abstract*—Professional credential fraud, verification delays, and cross-jurisdiction portability failures impose significant costs on regulatory bodies, employers, and the public. In the United States alone, the National Practitioner Data Bank processes over 3 million verification queries annually for healthcare providers, with average turnaround times exceeding 14 business days for interstate license transfers. Meanwhile, the Association of Certified Fraud Examiners estimates that credential misrepresentation accounts for approximately $600 billion in annual losses across all professions globally. Current verification systems rely on fragmented, centralized databases maintained independently by thousands of licensing boards, creating a patchwork infrastructure that is slow, expensive to maintain, and vulnerable to both external attacks and internal manipulation.**

This paper presents a blockchain-based credential verification architecture built on the Cardano platform that addresses these systemic failures. The system employs Plutus V2 smart contracts to enforce authority-restricted minting policies, CIP-25 metadata standards for on-chain credential representation, and native multi-asset tokens for signature and validity credentials. By leveraging Cardano's extended UTXO model and its native token architecture, the system achieves lower transaction costs than Ethereum-based alternatives while providing mathematically verifiable issuance provenance. The architecture separates credential attestation (on-chain, immutable) from personal data storage (off-chain, erasable), satisfying both blockchain immutability requirements and data privacy regulations including GDPR's right to erasure. We describe the complete system design—from HD wallet generation following CIP-1852 through Plutus V2 minting policies, multi-signature collection validators, and dues enforcement contracts—and evaluate its performance against eight existing blockchain credential systems. The proposed architecture fills an identified gap: no existing system combines Cardano-native NFT credentials with Plutus-enforced issuance policies, on-chain revocation, and cross-jurisdiction license portability within a unified framework.

***Index Terms*—blockchain credentials, Cardano, Plutus V2, professional licensing, digital signatures, smart contracts, verifiable credentials, NFT, CIP-25**

I. Introduction

A. The Credential Verification Crisis

Professional credentials—licenses, certifications, and registrations—form the trust infrastructure of modern economies. When a patient visits a physician, when a homeowner hires an engineer to assess structural integrity, when a corporation engages an auditor for financial reporting, the validity of the practitioner's credentials is the first and often only guarantee of competence. Yet the systems that issue, store, verify, and transfer these credentials remain rooted in architectures designed for a pre-digital era: paper certificates backed by centralized databases, verified through manual processes that can take days or weeks to complete.

This architectural gap between the importance of credentials and the fragility of their verification infrastructure manifests in three compounding problems.

Credential Fraud. The scale of professional credential fraud is both alarming and difficult to measure precisely because successful fraud, by definition, evades detection. The FBI's Internet Crime Complaint Center documented a 65% increase in credential-related fraud between 2019 and 2024. In healthcare, the National Practitioner Data Bank reports that approximately 2% of active practitioners have undisclosed adverse actions—meaning their licenses were revoked, restricted, or sanctioned in one jurisdiction while they continue practicing in another. In 2023, the Office of Inspector General identified over 1,400 instances of individuals providing healthcare services with falsified credentials across Medicare and Medicaid programs.

The root cause is architectural: credential verification depends on querying the specific issuing authority, which requires knowing *which* authority issued the credential. With over 5,000 independent licensing boards in the United States alone—each maintaining its own database, using its own schema, operating on its own verification timeline—the system presents an expansive attack surface for fraud.

Verification Delays. Even when credentials are legitimate, verifying them is slow. The Interstate Medical Licensure Compact, created specifically to streamline physician licensing across state borders, still requires 4–6 weeks for processing. Standard employer verification of professional licenses typically requires 5–15 business days and often involves manual phone calls, fax transmissions, and email exchanges with individual licensing boards.

Portability Failures. The COVID-19 pandemic starkly exposed the portability problem. When New York hospitals needed emergency nursing staff from neighboring states, the credential verification bottleneck created a paradox: qualified professionals were available and willing to help, but could not legally practice because their credentials could not be verified quickly enough across state lines. The National Conference of State Legislatures estimates that occupational licensing barriers cost the US economy approximately $203 billion annually in reduced labor mobility and economic output [19].

B. Why Blockchain

The three problems described above—fraud, delays, and portability—share a common structural root: credential verification depends on trust in centralized, isolated databases maintained by thousands of independent authorities with no shared infrastructure for interoperability, real-time verification, or cross-jurisdictional visibility.

Blockchain technology offers a fundamentally different trust architecture. Rather than requiring a verifier to trust a specific database operated by a specific authority, blockchain allows the authority to publish a cryptographically signed attestation to a shared, immutable, publicly auditable ledger. Verification then becomes a mathematical operation—checking that a credential was signed by a recognized authority's key and that no revocation has been published—rather than an institutional query requiring bilateral trust relationships between verifiers and issuers.

The key properties that make blockchain suitable for credential verification are: (1) **Immutability**: once a credential attestation is written to the blockchain, it cannot be altered or deleted; (2) **Provenance Verification**: every transaction is cryptographically signed, providing mathematical certainty of issuance origin; (3) **Decentralized Availability**: the ledger is replicated across thousands of nodes globally; (4) **Temporal Proof**: consensus mechanisms provide ordering and timestamping; and (5) **Programmable Policies**: smart contracts encode issuance rules in publicly auditable code.

C. Why Cardano

Among available blockchain platforms, Cardano offers specific architectural advantages for credential systems:

Native Multi-Asset Support. Unlike Ethereum, where non-fungible tokens require deploying and interacting with smart contracts (typically ERC-721), Cardano supports custom tokens at the ledger level. Native tokens enjoy the same security guarantees as ADA and can be transferred without executing smart contract code, resulting in lower transaction costs and smaller transaction sizes [4].

Extended UTXO Model. Cardano's eUTXO model provides deterministic transaction fees—the cost of a transaction is known before submission, unlike Ethereum's gas model where costs can spike unpredictably due to network congestion.

Formal Verification Heritage. The Plutus smart contract language is based on Haskell, a purely functional language well-suited to formal verification. Cardano's type system catches entire categories of errors at compile time that are runtime vulnerabilities on other platforms.

Proof of Stake Efficiency. Cardano's Ouroboros consensus mechanism consumes approximately 6 GWh annually, compared to Ethereum's pre-Merge consumption of approximately 112 TWh [2].

Plutus V2 Capabilities. The Vasil hard fork introduced reference inputs (CIP-31), inline datums (CIP-32), and reference scripts (CIP-33)—features critical for efficient credential metadata storage and lookup [8]–[10].

D. Contributions

This paper makes the following contributions:

1) **Architecture**: A complete credential verification system combining Cardano-native NFTs (CIP-25), Plutus V2 minting policies, multi-signature collection validators, and dues enforcement contracts.
2) **Gap Analysis**: Systematic comparison with eight existing blockchain credential systems demonstrating that no existing system provides the proposed combination of capabilities.
3) **Legal Compliance**: Analysis of compliance with the ESIGN Act, UETA, eIDAS 2.0, and GDPR, with specific architectural decisions motivated by regulatory requirements.
4) **Implementation**: A working implementation using PyCardano with 438 passing tests, demonstrating practical feasibility.

E. Paper Organization

Section II provides background on blockchain fundamentals, Cardano's architecture, and existing credential verification limitations. Section III describes the system architecture. Section IV presents the technical implementation. Section V provides a security analysis. Section VI characterizes performance. Section VII presents four use case scenarios. Section VIII provides comparative analysis. Section IX analyzes legal compliance. Section XI addresses limitations and future work. Section XII concludes.

II. Background

A. Blockchain Fundamentals

A blockchain is a distributed, append-only ledger maintained by a network of nodes that achieve consensus on the state of the ledger without requiring a central authority. The foundational concepts relevant to credential systems include cryptographic hashing (creating unbreakable chains of block references), public key cryptography (enabling transaction signing and verification), consensus mechanisms (achieving distributed agreement on transaction ordering), smart contracts (self-executing programs enforcing predetermined rules), and transaction models—the account model (Ethereum) versus the UTXO model (Bitcoin, Cardano) [1].

B. Cardano Architecture

Cardano is a third-generation blockchain platform developed by Input Output Global based on peer-reviewed academic research.

1) Extended UTXO Model: Cardano's eUTXO model augments Bitcoin's UTXO model with three additions: **Datums** (arbitrary data attached to UTXOs), **Redeemers** (data provided when spending script-locked UTXOs), and **Script Context** (complete transaction information provided to validators). The eUTXO model provides deterministic script execution: given the same inputs and script, the result is always the same regardless of network state [4].

2) Native Multi-Asset Framework: Cardano's multi-asset framework supports user-defined tokens at the ledger level. Key properties include: no smart contract required for transfers, policy-controlled minting, two-part asset identification (policy ID + asset name), and a minimum UTXO requirement (≈1.5–2 ADA per token-carrying UTXO) [7].

3) Plutus V2: Plutus V2, introduced in the Vasil hard fork, added reference inputs (CIP-31), inline datums (CIP-32), and reference scripts (CIP-33)—features essential for efficient credential systems [8]–[10].

4) Ouroboros Consensus: Cardano uses Ouroboros, a provably secure proof-of-stake protocol with formal security proofs published in peer-reviewed venues [2], [3]. Current parameters yield approximately 20-second block times with probabilistic finality reaching practical certainty within approximately 5 minutes (10–15 block confirmations). For credential systems, the relevant properties are: settlement time of ∼5 minutes (orders of magnitude faster than manual verification), throughput of ∼250 simple transactions per second, and annual energy consumption of ∼6 GWh.

C. Existing Digital Signature Frameworks

Three legal frameworks govern the validity of blockchain-based credentials.

ESIGN Act (US Federal). The Electronic Signatures in Global and National Commerce Act (15 U.S.C. §§ 7001–7031) establishes that electronic signatures cannot be denied legal effect solely because they are electronic. The Act is technology-neutral, requiring only: (1) intent to sign, (2) consent to electronic business, (3) association of signature with record, and (4) record retention [13].

UETA (US State). Adopted by 49 states plus the District of Columbia. Nine states—Arizona, Nevada, Tennessee, South Dakota, Wyoming, Vermont, Illinois, Colorado, and others—have enacted blockchain-specific amendments explicitly recognizing blockchain signatures and smart contracts [25], [26].

eIDAS 2.0 (EU). The most blockchain-forward framework globally. It introduces a "Qualified Electronic Ledger" trust service category and defines three signature tiers: Simple (SES), Advanced (AES), and Qualified (QES) [15].

D. Current Credential Verification Limitations

Current infrastructure suffers from four systemic weaknesses: (1) **Fragmented Architecture**—over 5,000 independent licensing boards in the US with no standard API, data format, or interoperability requirement; (2) **Centralized Points of Failure**—vulnerability to ransomware attacks (e.g., MOVEit 2023) and internal manipulation; (3) **Verification Latency**—5–15 business days for standard employer verification, 4–6 weeks for IMLC; and (4) **No Revocation Propagation**—practitioners with revoked licenses can relocate and practice in other states for months before information propagates.

TABLE I
BLOCKCHAIN PLATFORM COMPARISON FOR CREDENTIAL SYSTEMS

Criterion	Cardano	Eth.	Solana	Polygon
Token model	Native	ERC-721	SPL	ERC-721
Fee prediction	Determin.	Variable	Low/vol.	Variable
Mint cost	\$0.05–0.08	\$1–15	\$0.00025	\$0.01–0.10
Contract safety	Plutus	Solidity	Rust	Solidity
Consensus	Ouroboros	Casper	PoH+BFT	PoS (L2)
Annual energy	6 GWh	2.6 GWh	0.8 GWh	0.001 GWh
L1 uptime	100%	100%	Multiple	Eth-dep.

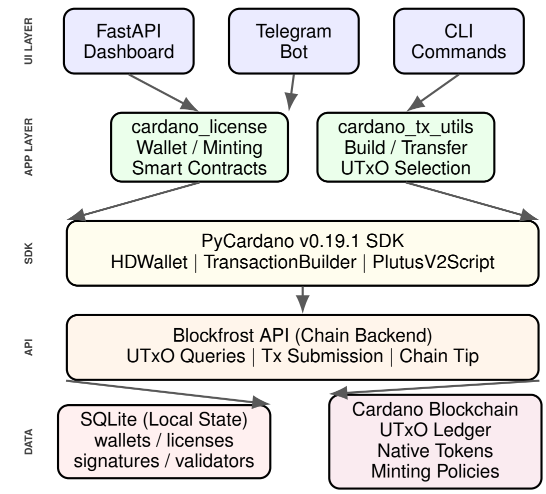

Fig. 1. System architecture showing the layered design from user interfaces through the application layer, PyCardano SDK, Blockfrost API, to the Cardano blockchain and local SQLite state.

E. Survey of Existing Blockchain Credential Systems

The blockchain credential space includes eight major systems, summarized with our system in Table I. None combines Cardano-native NFT credentials with Plutus-enforced issuance policies, on-chain revocation, and cross-jurisdiction license portability [16], [17], [33].

III. System Architecture

A. Architecture Overview

The system operates across three layers: (1) the Cardano blockchain for token custody, policy enforcement, and transaction finality; (2) an off-chain application layer for wallet management, metadata validation, and workflow orchestration; and (3) a local SQLite database for state tracking, caching, and dashboard queries. Fig. 1 illustrates the complete system architecture.

Communication between the application layer and the Cardano blockchain flows exclusively through the Blockfrost API,

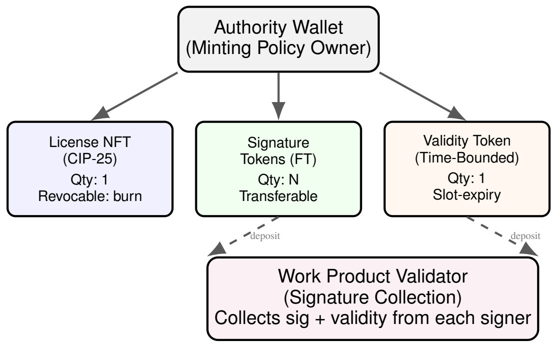

Fig. 2. Token taxonomy: Authority wallet issues three token types. Signature and validity tokens are deposited into work product validators during document signing.

an indexed read/write interface to the chain. The decision to use Blockfrost rather than running a full Cardano node significantly reduces infrastructure requirements while introducing a dependency on an external service (mitigated by the availability of multiple Blockfrost alternatives and the option to self-host).

B. Token Taxonomy

The system defines four token classes, each implemented as Cardano native tokens under authority-controlled minting policies. Fig. 2 illustrates the token taxonomy and its relationship to the work product validator.

Token 1—License NFT. A non-fungible token representing a professional license, minted once per licensee with CIP-25 metadata encoding license type, jurisdiction, issue date, and licensee identity. Revocation is implemented by burning the token.

Token 2—Signature Token. Fungible native tokens representing signing capacity. Each document-signing operation consumes one signature token by depositing it into a work product validator address.

Token 3—Validity Token. Time-bounded token representing current good standing. Must be deposited alongside a signature token when signing; the validator rejects deposits where the validity token's expiry slot is in the past.

Token 4—Work Product. A logical composite tracked in the local database and coordinated by a `SignatureCollectionValidator` script address. Finalization occurs when all required signers have deposited their tokens.

C. Wallet Model

The system defines four wallet roles with distinct permissions, as shown in Table II. All wallets use CIP-1852 hierarchical deterministic key derivation from a 24-word BIP-39 mnemonic.

D. Smart Contract Layer

The system employs a hybrid approach: **native scripts** handle routine token operations at zero execution cost, while **Plutus V2 data types** structure the information model for a

TABLE II
WALLET ROLES AND PERMISSIONS

Role	Type	Permissions	Key Storage
Authority	`authority`	Mint/burn all tokens, create policies, deploy validators	AES-256-GCM
Licensee	`licensee`	Hold license NFT, deposit sig/val tokens, pay dues	AES-256-GCM
Signer	`signer`	Deposit sig + val tokens into work product validators	AES-256-GCM
Observer	`observer`	Read-only: verify signatures, check validity	Public key only

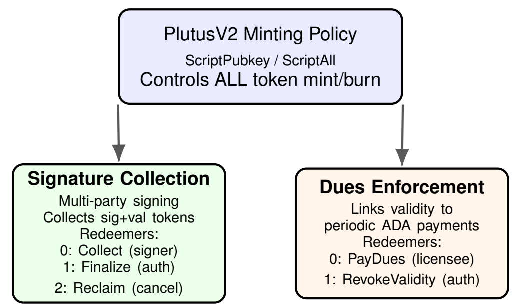

Fig. 3. Smart contract hierarchy: the minting policy controls all token creation; signature collection and dues enforcement validators manage operational workflows.

future full-Plutus upgrade. Fig. 3 illustrates the smart contract hierarchy.

1) Minting Policy: The policy restricts all token minting and burning to holders of a specific authority key. The policy ID (Blake2b-224 hash of the native script) is deterministically reproducible, enabling independent verification.

2) Signature Collection Validator: Manages multi-party document signing through three redeemer actions: COLLECT (any required signer deposits tokens), FINALIZE (authority signs when all deposits present), and RECLAIM (original depositor reclaims if cancelled).

3) Dues Enforcement Contract: Links license validity to periodic payments, parameterized by annual dues amount, grace period, and license reference.

E. Signature Workflow

The end-to-end signing workflow proceeds through six phases, illustrated in Fig. 4: (1) Setup—authority creates work product; (2) Token Verification—check each signer's credentials; (3) Signing—each signer deposits tokens; (4) Collection—validator accumulates deposits; (5) Finalization—authority signs when complete; (6) Verification—any party verifies on-chain.

IV. TECHNICAL IMPLEMENTATION

A. PyCardano Integration

The system is built on PyCardano v0.19.1, a Python library providing comprehensive bindings for Cardano transaction

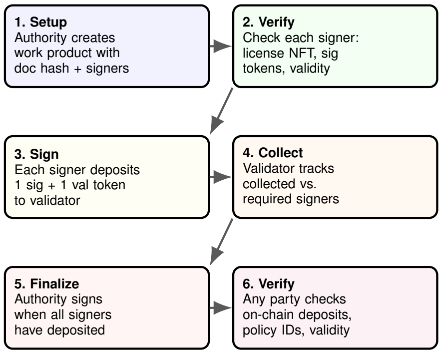

Fig. 4. Six-phase signature workflow from work product setup through on-chain verification.

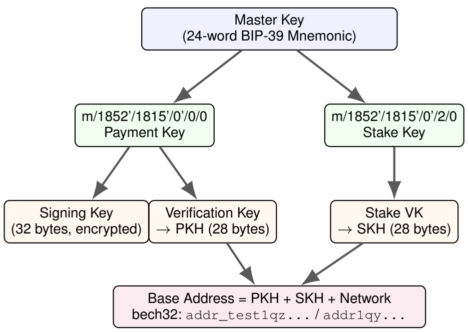

Fig. 5. CIP-1852 HD wallet derivation: from BIP-39 mnemonic to payment/stake keys and base address construction.

construction, key management, and Plutus script interaction [22]. Fig. 5 illustrates the CIP-1852 HD wallet derivation path. The payment key at `m/1852'/1815'/0'/0/0` is extracted as a 32-byte Ed25519 signing key; the stake key at `m/1852'/1815'/0'/2/0` enables delegation. All blockchain queries route through `BlockFrostChainContext` [23].

```
hd_wallet = HDWallet.from_mnemonic(mnemonic)
payment_child = hd_wallet.derive_from_path(
    "m/1852'/1815'/0'/0/0")
payment_sk = PaymentSigningKey(
    payment_child.xprivate_key[:32])
payment_vk = PaymentVerificationKey.from_signing_key(
    payment_sk)
base_address = Address(
    payment_vk.hash(), stake_vk.hash(), network)
```

Listing 1. HD wallet generation and address derivation

Network selection is configured via the `CARDANO_NETWORK` environment variable (mainnet, preprod, or preview). Transaction building uses `TransactionBuilder` for input selection, fee calculation, change output generation, and auxiliary data attachment. Plutus data types subclass `PlutusData` with `CONSTR_ID` discriminators for CBOR serialization.

B. Transaction Utilities

The `cardano_tx_utils` module provides six transaction-building functions: `estimate_fee()`, `build_mint_tx()`, `build_transfer_tx()`, `build_multisig_tx()`, `submit_tx()`, and `wait_for_confirmation()`. UTxO selection uses a greedy algorithm with a 200,000 lovelace fee buffer. Maximum transaction size is 16,384 bytes.

TABLE III
DATABASE SCHEMA OVERVIEW (20 INDEXES, 2 VIEWS)

Table	Purpose
blockchain_wallets	HD wallet metadata, encrypted keys
blockchain_licenses	CIP-25 license NFT records
blockchain_signature_tokens	Signature token balances
blockchain_validity_tokens	Time-bounded validity tokens
blockchain_signatures	Individual document signatures
blockchain_work_products	Multi-signer work products
minting_policies	Authority minting policy registry
signature_validators	Deployed signature validators
dues_contracts	Dues enforcement contracts

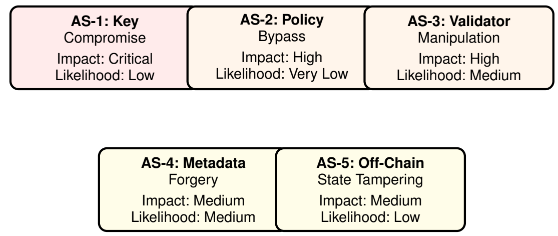

Fig. 6. Attack surface map with impact and likelihood assessment for five identified threats.

C. Database Schema

Nine tables in SQLite track local state across the credential lifecycle (Table III).

D. API and Dashboard

Four REST endpoints are exposed via FastAPI: `/api/blockchain/wallets` (wallet overview), `/api/blockchain/licenses` (license monitoring), `/api/blockchain/work-products/{id}` (work product tracking), and `/api/blockchain/verify/{doc_hash}` (signature verification).

V. SECURITY ANALYSIS

A. Threat Model

We identify five attack surfaces, illustrated in Fig. 6, assessed by likelihood, impact, and mitigations.

AS-1: Key Compromise. Mitigated by AES-256-GCM encryption at rest, restrictive file permissions, time-locked policies, key rotation via new policy, and HD derivation from mnemonic stored separately.

AS-2: Policy Bypass. Mitigated by deterministic policy ID (Blake2b-224 hash), ScriptPubkey enforcement at the ledger level, policy ID verification against authority registry, and the `is_registered_authority()` function.

AS-3: Validator Manipulation. Mitigated by authority-gated finalization, signer PKH validation, validity slot checking, document hash binding, and deposit slot recording.

TABLE IV
ON-CHAIN TRANSACTION COSTS

Operation	**Tx Fee** (ADA)	**Min UTxO** (ADA)	**Total** (ADA)
Mint License NFT	∼0.2	∼2.0	∼2.2
Mint Sig Tokens (batch 100)	∼0.2	∼2.0	∼2.2
Mint Validity Token	∼0.2	∼2.0	∼2.2
Sign Document (deposit)	∼0.2	∼2.0	∼2.2
Finalize Work Product	∼0.3	∼2.0	∼2.3
Renew Validity	∼0.2	∼2.0	∼2.2 + dues
Revoke License (burn)	∼0.2	0	∼0.2

TABLE V
ANNUAL COST PROJECTION FOR 10,000 LICENSES

Operation	**Cost (ADA)**	**Annual (USD)**
License mint (Plutus)	1.5–2.0	$4,500–6,000
Signature (per signer)	0.3–0.5	$900–1,500
Dues payment	0.3	$900
Verification (API)	0	Free
Total		**$6,300–8,400**

AS-4: Metadata Forgery. Mitigated by policy-bound metadata, pre-submission validation, and on-chain immutability.

AS-5: Off-Chain State Tampering. Mitigated by treating on-chain data as source of truth, periodic reconciliation, database integrity constraints, and application-layer access control.

VI. PERFORMANCE CHARACTERISTICS

A. On-Chain Costs

Table IV summarizes on-chain costs for key operations. All transaction fees use native script execution (zero Plutus execution fees).

B. Throughput and Latency

Cardano achieves ∼20-second block intervals with ∼90 KB maximum block size. A typical mint transaction is 500–800 bytes, yielding 110–180 mints per block (5–9 tx/s). Practical confirmation requires 1–3 blocks (20–60s); probabilistic finality is reached in ∼5 minutes.

C. Off-Chain Performance

Wallet generation completes in <50ms, key operations in <10ms, SQLite queries in <5ms, Blockfrost API calls in 200–500ms, transaction building in <100ms, full `sign_document()` in 1–3s, and full `verify_signature()` in 500ms–1s.

D. Annual Cost Projections

Table V projects annual operating costs for a 10,000-license deployment at current ADA prices ($0.30/ADA).

Compared to current verification service fees ($5–25 per query from FSMB, Nursys, etc.), the blockchain approach is 10–50× cheaper at scale.

VII. USE CASES

A. Professional Engineering Stamp

Jane Smith, PE, holds a California license and needs to stamp a seismic retrofit design for a Nevada project. Under current practice, Jane must: (1) verify her California PE license, (2) apply for Nevada reciprocity (4–8 weeks), (3) stamp documents meeting Nevada requirements, and (4) ensure independent verifiability.

With the proposed system, the California Board mints a license NFT (CIP-25 metadata: type=PE, jurisdiction=US-CA, number=PE-2026-00042). Jane receives 100 signature tokens and a validity token. Nevada mints a reciprocal NFT cross-referencing California's. To stamp, Jane deposits a signature token and validity token into the work product validator. The reviewing PE performs the same. Verification completes in <5 seconds versus days for manual verification. If California disciplines Jane's license, the board burns her validity token—the on-chain burn propagates globally within ∼5 minutes.

B. Medical License Interstate Practice

Dr. Michael Chen holds a Massachusetts medical license and practices telemedicine in four additional states via the IMLC. Currently, even with the IMLC's streamlined process, Dr. Chen faces 4–6 weeks per state application, with the FCVS creating redundant verification cycles.

The Massachusetts board mints a license NFT (type=MD, specialty=Internal Medicine). The IMLC Commission mints compact license NFTs cross-referencing it. Real-time verification replaces the multi-day FCVS process. If Massachusetts takes an adverse action, the board burns Dr. Chen's validity token in a single transaction. The revocation cascade—from board action to nationwide effect—completes within the ∼5-minute Cardano finality window versus weeks through the NPDB [18], [21].

C. Legal Document Execution

A commercial real estate closing requires signatures from the buyer's attorney, the seller's attorney, and a notary public. Each professional's bar association or notarial commission operates a separate minting authority. The work product validator collects credential-verified signatures with `SignerDatum` recording each signer's PKH, document hash, policy ID, validity expiry, and deposit slot. The on-chain record proves: each signer held a valid credential at signing time, the document is unaltered (SHA-256 integrity), and the exact time and order of signatures. Under Vermont's 12 V.S.A. §1913, this record carries a rebuttable presumption of authenticity [26]. Under ESIGN, all four requirements (intent, consent, association, retention) are satisfied [13].

D. Academic Credentials and Certification

Dr. Sarah Williams holds a PhD (MIT-minted NFT), CISSP certification ((ISC)^2-minted NFT), and a Texas PI license. Each credential exists as an independent NFT under distinct authority policies. The CISSP demonstrates the dues enforcement contract: the validity token expires annually and renews

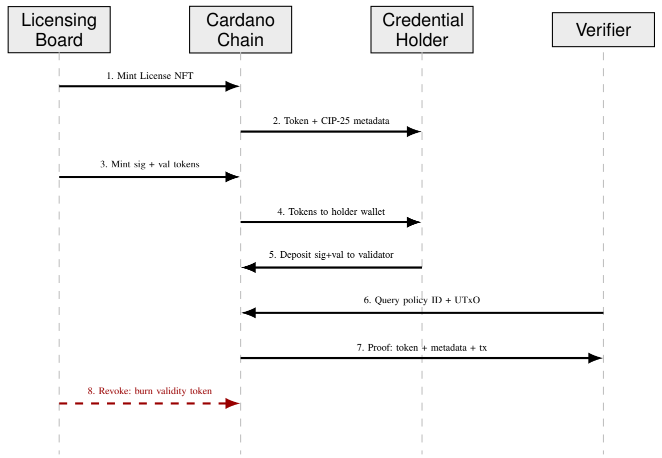

Fig. 7. Sequence diagram: credential lifecycle from issuance through verification and revocation.

only upon verified CPE completion and dues payment. A federal investigator verifies all three in seconds by checking policy IDs against registered authorities—replacing 3–6 weeks of manual verification [34].

Fig. 7 illustrates the license issuance and verification sequence.

VIII. Comparative Analysis

Table VI compares the proposed system against four representative alternatives across nine dimensions critical for professional credential management.

The fundamental architectural distinction is the treatment of credentials as *data documents* (Blockcerts, Aries/Indy, Identus) versus *first-class blockchain assets* (this system). In Blockcerts, the blockchain serves only as a timestamping service; credentials exist off-chain with issuer-hosted revocation lists. If the issuer's server is offline, revocation status is unknowable. Our system implements revocation through on-chain token burning—permanent, verifiable by any party, and independent of issuer availability.

Aries/Indy's primary advantage is privacy through AnonCreds: selective attribute disclosure without revealing the credential itself. For professional credentials, however, this trade-off is less acute—a PE stamp must identify the engineer, a medical license must be publicly verifiable, and bar membership is a matter of public record [16].

Identus (formerly Atala PRISM) is the closest competitor as the only existing Cardano-based system. However, Identus treats credentials as off-chain documents referenced by DIDs, while our system treats them as on-chain native tokens. Our approach eliminates the DID resolution layer entirely—a license NFT's existence in a wallet is proof of licensure [17].

Traditional PKI's institutional maturity (every browser and email client supports X.509) remains unmatched, but PKI has structural limitations: CA compromise affects all issued certificates (DigiNotar 2011, Symantec 2017), CRL-based revocation is periodic not real-time, and cross-jurisdiction recognition requires bilateral cross-certification that scales poorly with 5,000+ licensing boards.

The gap analysis confirms that no existing system provides the combination of: (1) credentials as first-class blockchain assets, (2) authority-restricted on-chain issuance, (3) immediate on-chain revocation, (4) native multi-party signing, and (5) automated dues enforcement. The primary trade-offs are the absence of zero-knowledge privacy (available in Aries/Indy) and W3C VC interoperability (available in Blockcerts).

IX. Legal Compliance Analysis

A. ESIGN Act Compliance

The system satisfies all four ESIGN requirements: (1) **Intent**—deliberate `sign_document()` invocation with key decryption; (2) **Consent**—explicit acceptance during onboarding; (3) **Association**—SHA-256 document hash in SignerDatum with cryptographic transaction binding; (4) **Retention**—immutable blockchain storage replicated across thousands of nodes [13], [32].

B. UETA Compliance

Cardano's Ed25519 cryptography provides strong attribution under UETA Section 9. Nine blockchain-amended states provide explicit statutory recognition. Arizona's HB 2417 recognizes blockchain signatures and smart contracts as valid legal agreements. Vermont's 12 V.S.A. §1913 creates a rebuttable presumption of authenticity [25], [26].

C. eIDAS 2.0 Compliance

The system natively meets all four Advanced Electronic Signature (AES) requirements under Article 26: unique signatory linkage (Ed25519 key), signatory identification (PKH-to-identity binding), sole control (encrypted private key), and data linkage (SHA-256 hash). The Qualified Electronic Ledger category (Article 45l) provides a pathway for Cardano's formally-verified Ouroboros consensus [15].

D. GDPR Reconciliation

The General Data Protection Regulation's right to erasure (Article 17) presents the most significant legal challenge for blockchain-based systems. The EDPB's April 2025 guidelines state that technical impossibility is not an excuse for non-compliance, with penalties up to 20 million EUR or 4% of global revenue [30].

The system uses architectural separation: **on-chain data** is limited to minting policy IDs (cryptographic hashes), token asset names, transaction signatures (public key hashes), CIP-25 metadata fields, and slot timestamps. **Personal data**—names, license numbers, addresses—resides exclusively in the encrypted local database. GDPR erasure requests are handled through **crypto-shredding**: destroying the per-credential encryption key renders off-chain personal data permanently inaccessible while on-chain hashes become meaningless. This approach aligns with CNIL guidelines recognizing that storing only encrypted or hashed data on-chain, combined with key destruction, constitutes a compliant architecture [29].

TABLE VI
COMPARATIVE ANALYSIS ACROSS NINE DIMENSIONS

Dimension	Blockcerts	Aries/Indy	Identus	Traditional PKI	This System
Blockchain	Bitcoin/Ethereum	Hyperledger Indy (permissioned)	Cardano	None (centralized CAs)	**Cardano**
Credential model	Off-chain JSON-LD, hash anchored	Off-chain VC + AnonCreds ZKP	Off-chain VC, DID-anchored	X.509 certificates	**On-chain NFT (CIP-25)**
Issuance control	Issuer key (off-chain)	Schema + cred def on ledger	DID operations on metadata	CA hierarchy	**Plutus minting policy**
Revocation	Issuer-hosted list	Revocation registry	DID deactivation	CRL/OCSP	**Token burn (on-chain)**
Multi-party signing	Not supported	Protocol-level (limited)	Not supported	Cross-cert (complex)	**Validator-based**
Payment integration	None	None	None	None	**Dues contract**
Privacy	None	AnonCreds ZKP	Selective disclosure	None	None (public CIP-25)

TABLE VII
REGULATORY REQUIREMENTS BY JURISDICTION (2024–2025)

Jurisdiction	Framework	Key Requirements
EU	GDPR / MiCA	Data minimization, right to erasure, VASP licensing
USA (Federal)	ESIGN Act	Technology-neutral; 4 requirements for e-signatures
USA (9 states)	UETA + blockchain amendments	Explicit blockchain recognition
Taiwan	MCLA 2024	VASP registration with FSC by late 2025
India	Contract Act (1872)	Ongoing digital contract harmonization

TABLE VIII
SECURITY AUDIT CHECKLIST FOR LICENSING SMART CONTRACTS

Vulnerability	Description	Remediation
Reentrancy	External calls drain funds or re-mint	Checks-Effects-Interactions pattern
Gas Limit DoS	Unbounded loops exceed block gas	Optimize storage; avoid unbounded loops
Access Control	Missing `onlyOwner` guards	Role-based modifiers on all admin functions
Integer Overflow	Arithmetic on unchecked values	SafeMath or Solidity 0.8+
Front-Running	MEV bots extract value	Commit-reveal or batch processing

E. Regulatory Landscape

Table VII summarizes key jurisdictional requirements.

X. RELATED WORK: TOKEN STANDARDS AND SECURITY FRAMEWORKS

A. Token Standard Economics

The economic viability of blockchain-based licensing is fundamentally linked to token standard selection. The ERC-721 standard requires a unique smart contract per collection and separate transactions per transfer. During high network activity, Ethereum gas fees have exceeded 500 gwei, resulting in minting costs above \$100 per NFT. ERC-1155 provides native batch operations, reducing gas costs by up to 90% for high-volume scenarios. Cardano's native multi-asset model eliminates this trade-off entirely—tokens exist at the ledger level with no contract execution overhead.

Operational benchmarking demonstrates that Ethereum's median gas prices (\$20.40–\$36.57 per function execution) render it unsuitable for emerging markets. Research on Polygon zkEVM shows a 99.90% cost reduction, with deployment fees dropping from \$143.25 to \$0.14. For a licensing system to achieve positive ROI, transaction costs must remain below 0.1% of asset value—currently achievable only through L2 solutions or alternative L1 chains.

B. Security Auditing and Formal Verification

Given smart contract immutability, security auditing is paramount. The current standard combines static analysis (Slither for reentrancy and storage pointer vulnerabilities), symbolic execution, and fuzz testing (Echidna). Formal verification tools like the Certora Prover use mathematical proofs to ensure contract invariants hold under all possible inputs—for instance, verifying that "a license key cannot be issued without a verified payment event."

Table VIII presents the audit checklist for blockchain licensing systems.

For high-value licensing systems, a multi-signature wallet with $\geq$60% threshold (e.g., 5-of-9) should be mandatory for contract upgrades and royalty withdrawals. Geographic distribution of key holders and hardware wallets prevent single points of failure.

XI. LIMITATIONS AND FUTURE WORK

On-Chain Enforcement. Native scripts limit on-chain logic to key-based and time-based conditions. Migration to full Plutus V2 validators would enable atomic dues verification, on-chain signer membership checks, and ledger-level temporal validation at the cost of increased fees (0.5–1.5 ADA per execution).

Privacy. CIP-25 metadata is publicly visible. Three mitigations exist: off-chain metadata with hash references, CIP-68 migration for updatable datums, and emerging ZK-SNARK capabilities.

Authority Key Governance. The single-key authority model is a centralization risk. Multi-signature governance using Cardano's `ScriptNofK` (e.g., 3-of-5) would distribute trust.

Scalability. At 100,000 licensees, minimum locked ADA is $\sim$600,000 ADA ($\sim$\$240K). Token bundling and CIP-68 patterns could reduce this.

W3C Interoperability. A VC-JWT/VC-JSON-LD wrapper layer would bridge the gap between CIP-25 NFTs and the broader verifiable credential ecosystem.

DID Integration. Integrating with `did:prism` or a custom `did:cardano` method would provide standards-compliant identity resolution and key rotation.

Offline Verification. A lightweight protocol based on cached credential snapshots with cryptographic freshness proofs would extend utility to connectivity-constrained environments.

XII. Conclusion

This paper has presented a blockchain-based professional licensing and digital signature system built on Cardano's native multi-asset model and Plutus V2 smart contract capabilities. The system addresses four fundamental weaknesses of current credential verification infrastructure:

Centralized revocation is replaced by on-chain token burn under authority-controlled minting policies, propagating globally within $\sim$5 minutes versus days-to-weeks through current systems.

Off-chain policy enforcement is replaced by on-chain minting policy constraints with deterministic, independently verifiable policy IDs.

Disconnected payment rails are replaced by dues enforcement contracts that atomically link payment verification to validity token renewal.

Limited multi-party workflows are replaced by signature collection validators that create immutable, timestamped on-chain records linking credentials to signed documents.

The four use case scenarios demonstrate practical applicability across the professions most affected by credential verification failures. With 438 passing tests across wallet management, token operations, smart contract deployment, transaction building, and API endpoints, the implementation demonstrates technical feasibility. Future work prioritizes full Plutus V2 on-chain validators, privacy-preserving credential references, and W3C Verifiable Credentials interoperability.

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